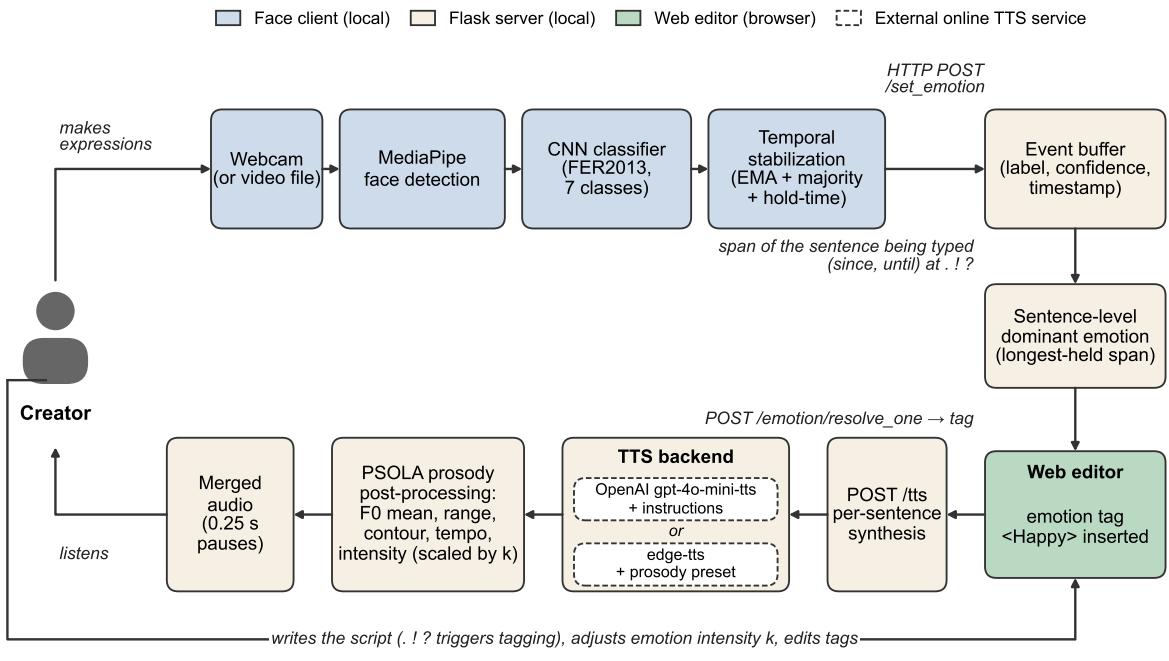
Face client (local)
Flask server (local)
Web editor (browser)
External online TTS service
HTTP POST
/set_emotion
makes
expressions
Webcam
(or video file)
MediaPipe
face detection
CNN classifier
(FER2013,
7 classes)
Temporal
stabilization
(EMA + majority
+ hold-time)
Event buffer
(label, confidence,
timestamp)
span of the sentence being typed
(since, until) at . ! ?
Sentence-level
dominant emotion
(longest-held span)
Creator
POST /emotion/resolve_one → tag
Web editor
emotion tag
<Happy> inserted
POST /tts
per-sentence
synthesis
TTS backend
OpenAI gpt-4o-mini-tts
+ instructions
or
edge-tts
+ prosody preset
PSOLA prosody
post-processing:
F0 mean, range,
contour, tempo,
intensity (scaled by k)
Merged
audio
(0.25 s
pauses)
listens
writes the script (. ! ? triggers tagging), adjusts emotion intensity k, edits tags